Enhancing Real Estate Business Through Data Driven Analysis

BDM CAPSTONE PROJECT FINAL SUBMISSION

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(BS) DEGREE IN DATA SCIENCE AND APPLICATIONS

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Executive Summary

Proprietor: Mr. Rakesh Narang

Organization: Narang Associates

Site: S-17B, Tent Market, Indira Nagar, Aadarsh Nagar, Delhi-110033

Employees working: 4

Operating Hours: 10:00 AM - 7:00 PM, Monday to Saturday

Years in Operation: 10 Years

Narang Associates, a small real estate business in Indira Nagar and Aadarsh Nagar, Delhi, faces significant challenges due to limited market reach and intense competition. This project aims to provide actionable insights to overcome these obstacles by employing data-driven analysis. The primary objective is to enhance market positioning, expand the business into more profitable regions, and improve decision-making through data insights.

Key methods used include descriptive statistics to analyze transaction data, inferential statistics for identifying trends, and correlation analysis to uncover relationships between customer demographics and deal types. Visual tools such as bar graphs, pie charts, and trend lines were applied to represent market dynamics clearly.

The analysis revealed that Narang Associates could significantly improve profitability by shifting focus to higher-commission areas, such as more developed regions with rising property rates. Additionally, the business can differentiate itself in a competitive market by offering unique services, exploring under-construction projects, and targeting niche markets. Recommendations include expanding into new geographic areas, optimizing marketing efforts, and partnering with developers to capitalize on upcoming property trends.

These insights are expected to help Narang Associates overcome current challenges, increase market reach, and improve profitability in the near future. Through a more strategic approach, the business can ensure sustained growth and better positioning in the highly competitive real estate market.

Detailed Explanation of Analysis Process

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1. **Data Collection**: Data was collected from interviews with the owner of Narang Associates and business records. Key variables included property type (shop, flat, plot), deal type (sale or rental), property location, reason for buying (commercial or residential), deal amount, deal source, commissions earned from buyer and seller, client gender, client age, number of dealers involved, and expenses incurred. Additionally, Google Maps data was used to assess the competitive landscape by mapping the number of real estate agents operating in the same areas, giving insight into market saturation.

2. **Data Cleaning**: The initial data was unstructured and handwritten. Issues included missing values, lack of detailed client information, and unclear records on business operations. Using Excel tools, the data was cleaned and structured. This process involved organizing property deals by location, standardizing commission details, and clarifying client demographics.

3. **Descriptive Statistics**: Descriptive statistics helped summarize the data with key metrics such as the average commission per deal, median deal size, and the frequency of property type transactions (plots, flats, shops). A breakdown of deals by location identified high and low-performing areas, which were visually represented through bar and pie charts, offering clear trends on deal types and commissions.

4. **Competitive Analysis**: Data from Google Maps was used to conduct a competitor analysis. Screenshots from Google Maps showed the proximity and density of competing real estate agents near Narang Associates' locations. A competitor density map was created to visually represent how saturated the market is, providing insight into the competitive pressures faced by Narang Associates.

5. **Correlation Analysis**: A correlation analysis was conducted to examine the relationship between customer age and deal type (sale or rent). The analysis revealed that middle-aged customers (33-45 years) were more inclined towards high-value property sales. This insight helped guide targeted marketing strategies to capture higher-commission deals.

6. **Trend Analysis**: An income trendline was plotted on a bar chart that showed deals per location, indicating the profitability trends over time. This analysis revealed which locations were generating increasing profits, enabling the business to focus on more profitable areas. Additionally, seasonality patterns in sales and rentals were identified to predict future fluctuations in revenue.

7. **Geographic Market Expansion**: The analysis suggested expanding into more lucrative areas like South Delhi and Connaught Place, where property prices and commissions are higher. The potential of targeting under-construction and upcoming projects was also explored. A cost-benefit analysis was performed to evaluate profitability in these areas, helping Narang Associates strategize for future growth.

8. **Inferential Statistics**: A chi-square test was used to determine if location significantly impacted deal closure rates. Confidence intervals were also calculated for key metrics like average commission and deal amounts. This statistical testing provided a clearer picture of potential outcomes, enabling data-backed decisions for geographic shifts and deal types.

9. **Customer Segmentation**: Customers were segmented by age and income, revealing that younger customers (28-35 years) tended to rent, while those in the 40-50 range preferred to buy. This segmentation helped the business tailor its marketing strategies to different customer groups, ensuring targeted outreach and optimized resource allocation.

10. **Visualization Tools**: Bar charts and pie charts were used to represent the frequency of property deals, customer demographics, and competitor density. Stacked bar charts compared different property types and transaction types (sale vs. rent) across locations, while trendlines and scatter plots helped visualize income progression and the correlation between deal amounts and customer demographics.

11. **Conclusion of Analysis Process**: The analysis revealed that Narang Associates performs well in specific market segments, but there are untapped opportunities in higher-commission areas. Key recommendations include focusing on high-value property sales, expanding into new markets with more profitable potential, and differentiating the business through unique services or strategic partnerships. These actions are crucial to stand out in a highly competitive environment and drive further growth.

Results and Findings

Introduction: This section presents the key insights derived from the data analysis conducted on Narang Associates' real estate operations. Through the use of various analytical techniques such as bar charts, pie charts, and correlation graphs, the findings provide a detailed overview of deal patterns, customer demographics, and profitability by location and property type. The results highlight critical trends and opportunities for growth, offering actionable insights that can help Narang Associates refine their business strategies, increase market reach, and improve overall profitability.

Combined Insights from Both Graphs:

- **High-Performing Areas (Indira Nagar, Model Town, Adarsh Nagar)**: Both graphs consistently show that these locations are the primary sources of business. The high deal counts and large portions of the pie chart suggest that the business should continue focusing its efforts in these areas, perhaps by investing more in marketing or offering additional services to capitalize on their existing strength.
- **Potential Growth Areas (Azadpur and Majlis Park)**: These locations are performing moderately, and the business could look into what factors might be limiting growth here. Perhaps by introducing new marketing strategies, partnerships, or focusing on customer satisfaction, the business could increase its presence in these areas.
- **Underperforming Areas (Mukundpur and Kewal Park)**: Both graphs highlight that these locations are underperforming. The business could explore why these areas are generating fewer deals—whether it's due to less demand, fewer marketing efforts, or tougher competition. Special attention could be given to these regions to improve their performance, such as targeting niche markets or offering special deal

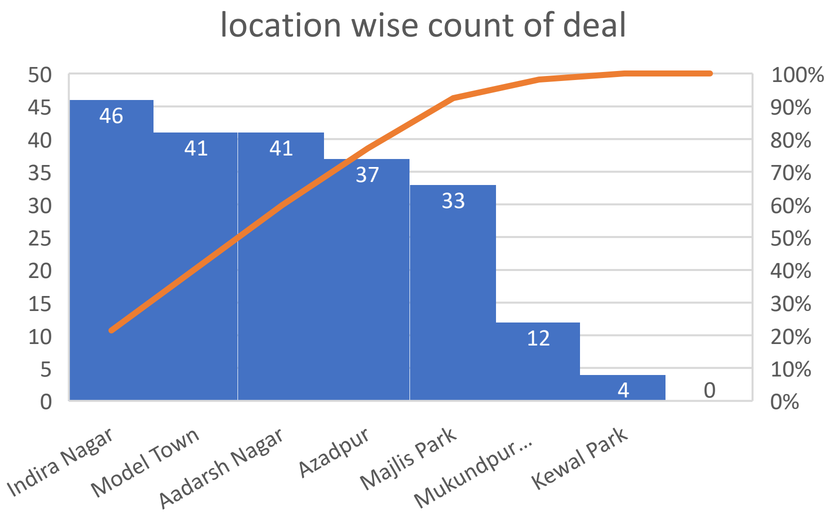


Figure 1:Bar Graph shows number deals per location

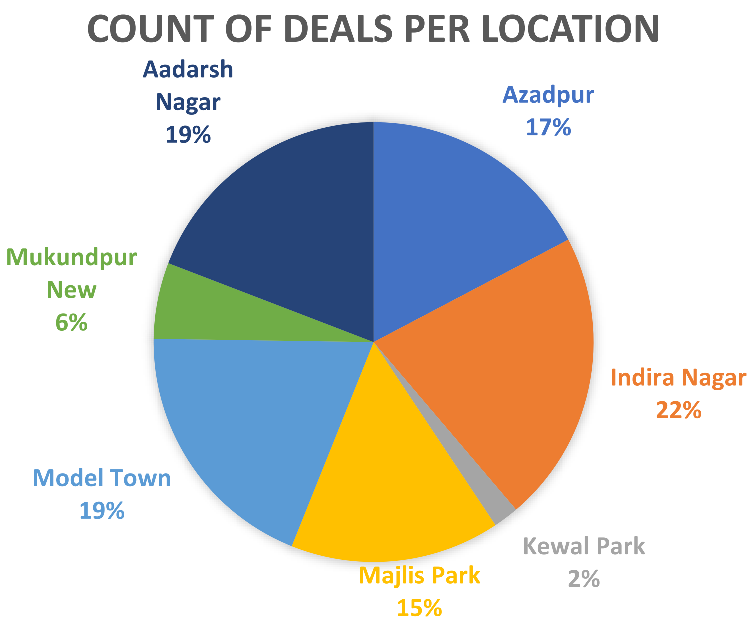


Figure 2:Pie chart shows location wise contribution

Location-Specific Performance and Strategic Focus:

1. **Azadpur**: With 0 sales, this may indicate low demand, ineffective marketing, or mismatched property offerings. We should explore rental opportunities or investigate reasons for the lack of sales, such as pricing or customer preferences.

2. **Kewal Park**: With equal sales and rentals, it shows balanced demand, making it suitable for targeting both buyers and renters. Seasonal trends should be analyzed to optimize strategy year-round.

3. **Other Locations**: High rental activity suggests strong demand, possibly due to demographics like students or professionals. To capitalize, we should enhance rental services and consider expanding rental property options.

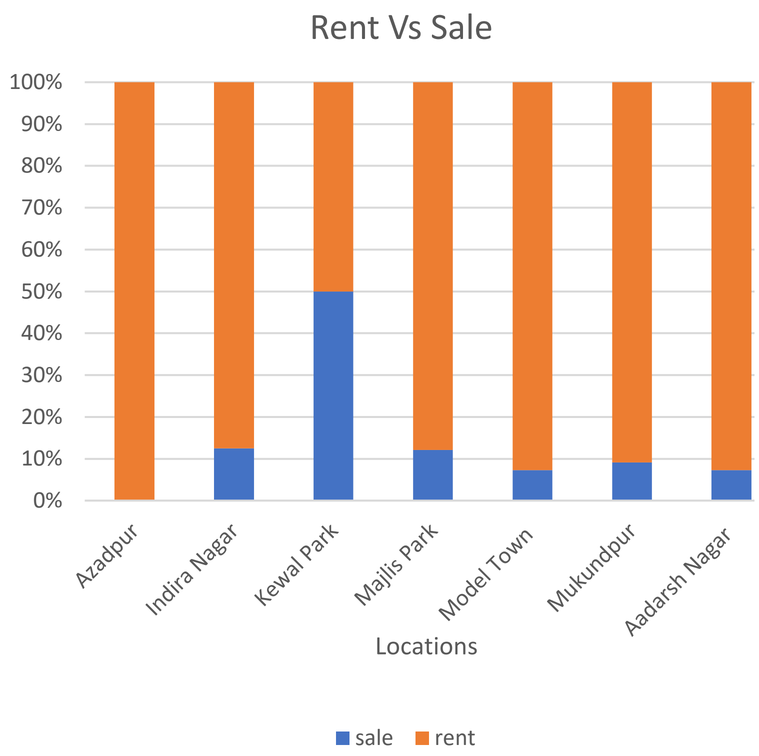


Figure 3:Stacked column chart showing proportions of rent and sales

Strategic Focus:

- Focus on rental services where rentals dominate, positioning the business as a rental specialist.

- For areas like Kewal Park with stronger sales, shift marketing efforts towards property sales.

Business Development:

- In underperforming locations like Azadpur, further market research is needed to identify barriers and improve sales or shift focus to more profitable areas.

Growth Opportunities:

- Invest in high-demand rental areas and consider partnerships with builders to expand rental offerings where supply is limited.

Key Insights:

- Azadpur requires improved sales strategies or market understanding.

- Kewal Park supports both sales and rentals, offering growth potential for both.

- Strong rental areas should prompt a shift toward enhancing rental services and property portfolios.

Total Deals by Location and Strategic Insights

1. **Total Deals by Location**:

    - **High Deal Volume Locations**: Indira Nagar accounts for 20% of total deals, making it the best-performing area. This suggests high demand and frequent transactions, making it a key focus for both sales and rentals.

    - **Low Deal Volume Locations**: Kewal Park has the fewest deals, suggesting that it may need a different strategy, such as enhanced marketing or a focus on other property types, to increase its performance.

2.Amount of Deals (Transaction Size):

- **Locations with High Deal Amounts**: Indira Nagar not only has the most deals but also the highest deal amounts, indicating that this area generates significant commission due to high-value transactions. Majlis Park follows closely, with fewer deals but still high-value properties that offer strong commissions.

- **Low Deal Amount Locations**: Aadrash Nagar and Model Town have many deals but at lower values. These areas are currently producing less commission, which suggests that while there is transaction volume, the properties being dealt with are likely lower in value. This could indicate a need to target higher-end properties or work on increasing commission rates. Kewal Park, with the lowest deal volume and value, is currently underperforming.

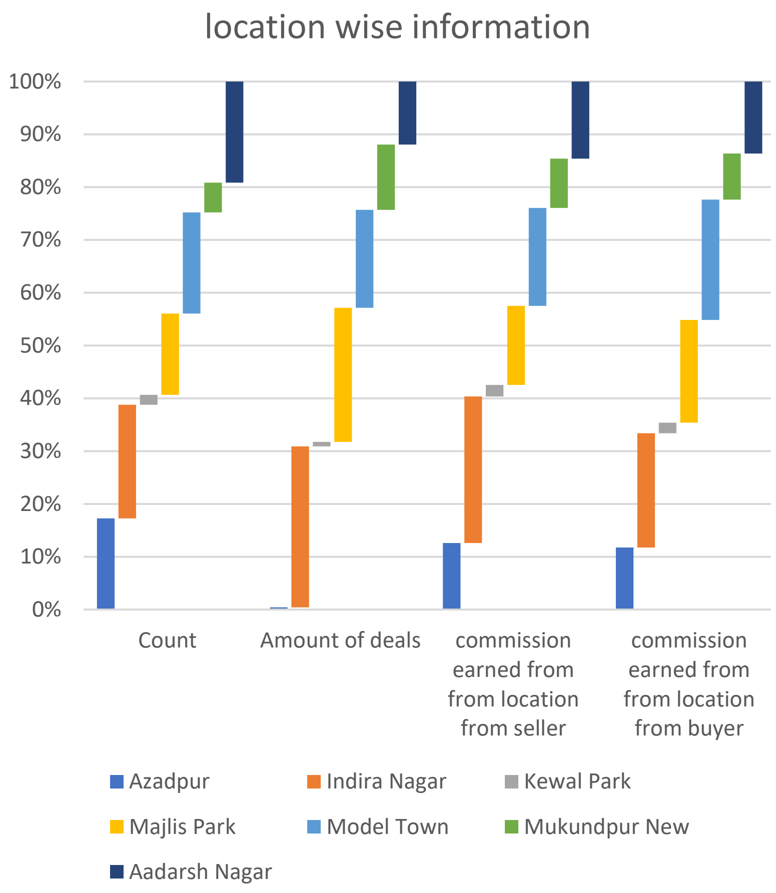


Figure 4:Stacked bar chart to show information location wise

Strategic Insights:

- **Focus on Indira Nagar and Majlis Park**: These two locations generate the highest returns in terms of both deal volume and transaction sizes. Business efforts should prioritize these areas to maintain strong relationships and further build a presence.

- **Optimize Aadrash Nagar and Model Town**: Despite high deal counts, the value of transactions in these areas is lower. A strategic focus on targeting more premium properties or renegotiating commission rates in these areas could improve profitability.

- **Reevaluate Kewal Park**: With the lowest deal count and transaction values, Kewal Park is contributing minimally to overall commission. The business may need to reconsider its focus on this area unless future potential is identified, or it might need a completely new approach, such as targeting rentals or alternative property types.

COMMISSION EARNED FROM RENT OR SALE:

Insights from these 3 graphs

1. **Location-wise Focus:**
    - **Azadpur:** The lack of commission from sales in Azadpur indicates a low or nonexistent sales market there. However, this could imply that Azadpur might be a prime area for rental properties, as we can focus on rental demand or adjust strategies to improve sales in this location.
    - **Indira Nagar:** The significantly higher commission from sales compared to rent in Indiranagar shows that this location has a strong sales market. This suggests that we can focus on promoting property sales in Indiranagar, as it's performing well in that regard.
    - **Kewal Park:** With the lowest commission from both sales and rent, Keval Park may not be as profitable. We could either reconsider investing efforts in this area or explore ways to boost market activity, possibly by collaborating with other agents, offering discounts, or marketing property benefits more effectively.

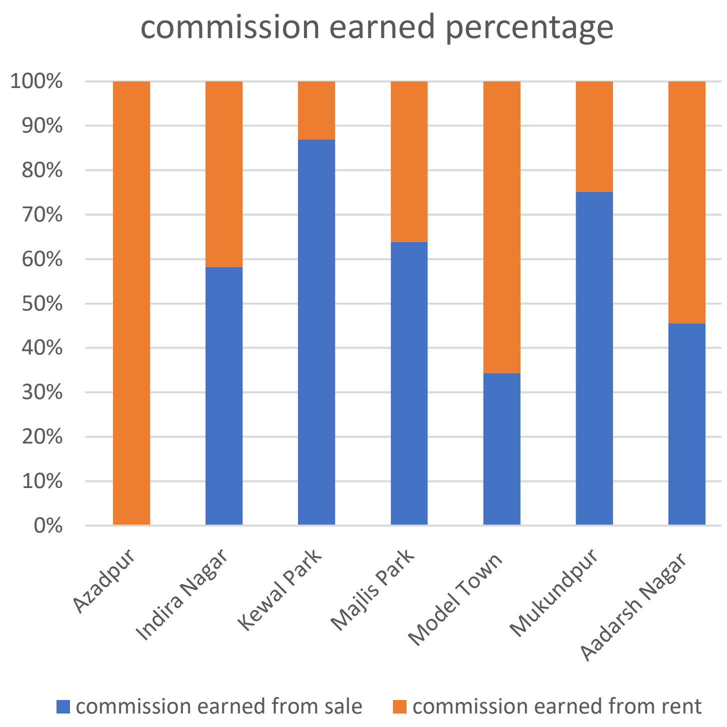


Figure 5:Stacked column chart showing commission earned

2. **Commission Balance:**
    - **Majlis Park:** The commission is fairly balanced, though leaning more towards sales. This suggests that both sales and rentals perform well in this location, making it a versatile area for both types of deals. We could take advantage of this by promoting it to a broad audience, including investors and renters.
    - **Model Town:** The higher commission from rent (120,000) versus sales (60,000) suggests that Model Town is more of a rental hotspot. We may want to focus marketing efforts on renters here, emphasizing affordability and availability of rental properties.
    - **Mukundpur and Adarsh Nagar:** Both locations show relatively balanced commissions between sales and rent, with slight differences. This indicates that these areas are equally viable for both sales and rentals, and strategies targeting both markets would likely yield returns.

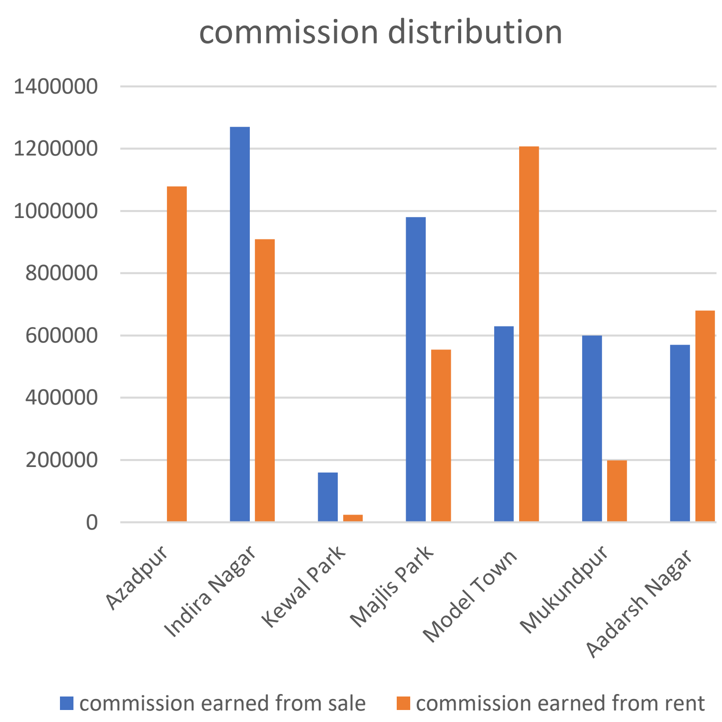


Figure 6:Stacked column chart showing commission distribution

Recommendations:

1. **Location-Specific Strategies:**
   - For **Azadpur**, we can focus more on rental properties, as the sales market seems weak. Increase listings of rental properties or emphasize rental options in your marketing campaigns for this area.
   - For **Indiranagar**, continue promoting property sales, as this is where you're generating the most commission. We might also offer competitive incentives for buyers to further boost sales.
   - For **Keval Park**, investigate the reasons for low commissions in both sales and rent. We may need to re-strategize or investigate potential opportunities that haven't been explored yet.
2. **Rental vs Sales Emphasis:**
   - In locations like **Model Town**, with a stronger rental market, tailor marketing to renters—highlight properties that are in demand for rentals and consider partnerships with rental-focused businesses.
   - In locations like **Indiranagar** and **Majlis Park**, where sales are strong, direct business efforts towards attracting buyers, including offering financing options, partnerships with real estate investors, or targeted advertisements that appeal to buyers.
3. **Portfolio Diversification:**
   - The data suggests that having a balance between sales and rental properties is key to maximizing commission. In locations with an equal mix of both sales and rental commissions (e.g., Adarsh Nagar, Mukundpur), we can maintain flexibility and market both types of properties to optimize revenue.
4. **Revenue Projections:**
   - Use these trends to predict future revenue streams. For instance, if the rental market is consistently stronger in certain locations, we may want to prioritize investment in rental properties there.
   - Similarly, locations like **Indiranagar** and **Majlis Park** can be forecasted for higher sales activity, guiding resource allocation and marketing budgets accordingly.

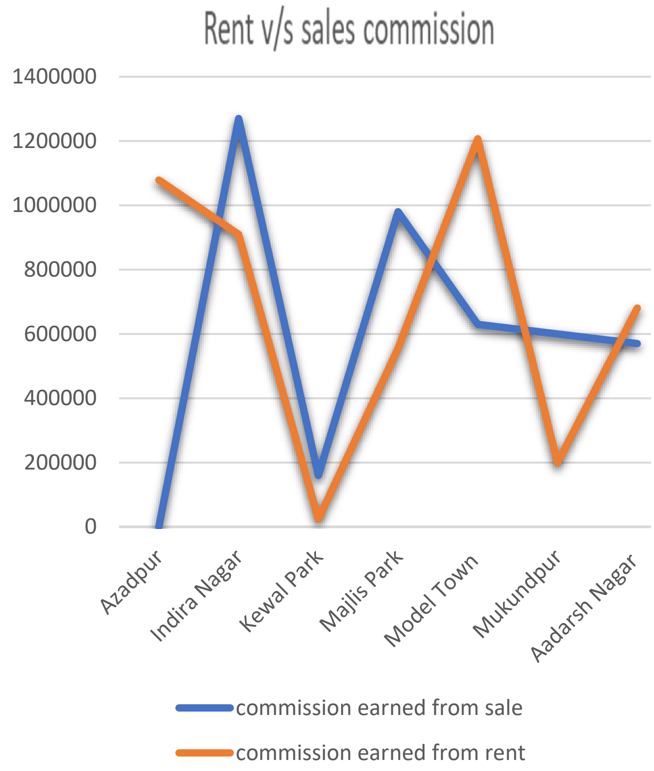


Figure 7: Line chart showing trend line in commission earned through sale and rent

MARKET SEGMENTATION:

Analysis:

1. **Target Market Segmentation**:
    - The data indicates that your key demographic is between 28-48 years of age, especially those under 33. This could help in tailoring marketing strategies to younger professionals or families.
    - The decline in older age groups suggests that you may not need to target these segments as heavily for property sales, though there may be opportunities for rental or investment-based properties for older clients.
2. **Product Offering Adjustment**:
    - Since younger clients are more likely to purchase, you might consider focusing more on offering properties that appeal to first-time homebuyers, such as affordable flats or houses in locations with good infrastructure and employment opportunities.
    - For the slightly older group (48-53), there may be interest in second homes, investment properties, or retirement-friendly housing solutions.
3. **Lifecycle and Future Forecasting**:
    - As the current younger generation ages, we could expect a shift in demand for larger homes or family-oriented properties. Tracking these trends over time would allow you to anticipate future demand in different segments.
4. **Marketing and Sales Strategy**:
    - For younger buyers, focus on digital marketing campaigns and partnerships with financial institutions offering attractive loan options. Highlight the advantages of early property investment, and use customer testimonials from younger clients.
    - For older clients, consider marketing properties that cater to retirement, downsizing, or investment properties with rental potential, which could appeal more to those in their late 40s and 50s.

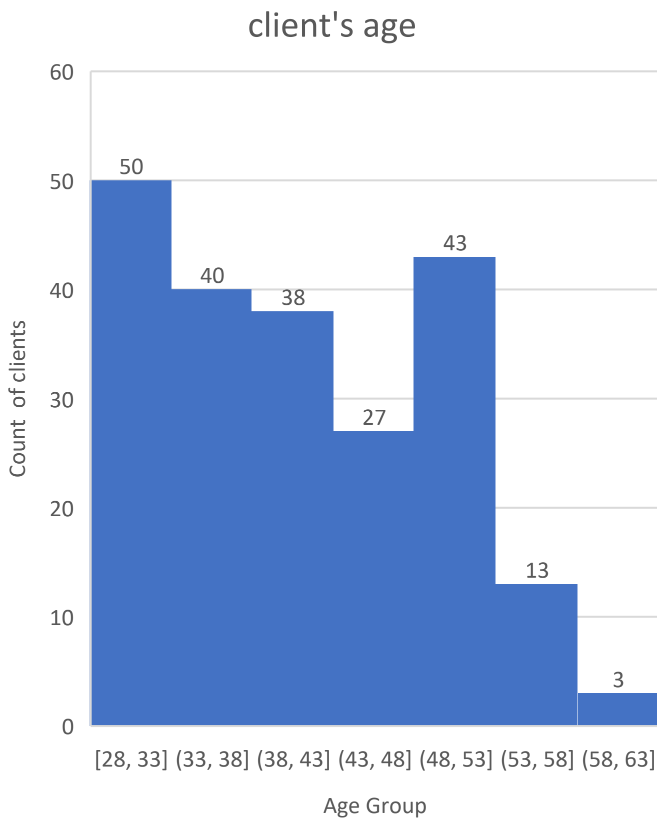


Figure 8:Histogram showing age of clients

ANALYSIS OF PROPERTY TYPE:

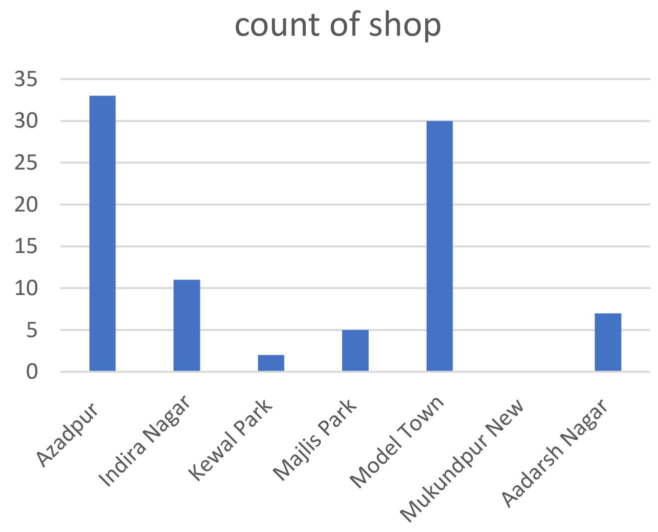


Figure 9:Bar graph showing count of shops

Analysis of Shops:

- **Azadpur** has the highest count of shops, indicating that it is likely a commercial hub where retail properties are in high demand. This could mean that businesses prefer this location for setting up shops, possibly due to higher foot traffic or accessibility.

- **Model Town** follows as the second most significant location for shops, which might also suggest a growing commercial presence but perhaps with less prominence compared to Azadpur.

- **Indira Nagar**, **Adarsh Nagar**, and **Majlis Park** also have a reasonable number of shops, suggesting they are important but secondary commercial areas.

- **Kewal Park** has the lowest count of shops, indicating that commercial activities are relatively low or that the area is more residential or underdeveloped commercially.

Insights for Shops:

- The high count of shops in Azadpur and Model Town suggests these areas could be key commercial centers, making them prime targets for investment or expansion in retail-focused real estate deals.

- Areas like Kewal Park might offer opportunities for future growth if the commercial demand increases. Business could explore the reasons for its lower commercial activity and assess potential for future development.

Analysis of Plots:

- **Majlis Park** has the highest count of plots (though still small at two), indicating limited but potentially valuable land sales.

- **Indira Nagar** has one plot, showing that plot sales are very minimal.

- No other locations show plot sales, indicating that plots are not a significant focus in these areas.

Insights for Plots:

- The scarcity of plots across most locations suggests that land deals are rare, and when available, they could represent high-value transactions due to limited availability.

- Areas like Majlis Park might have future growth potential if more plots become available or if development initiatives take place.

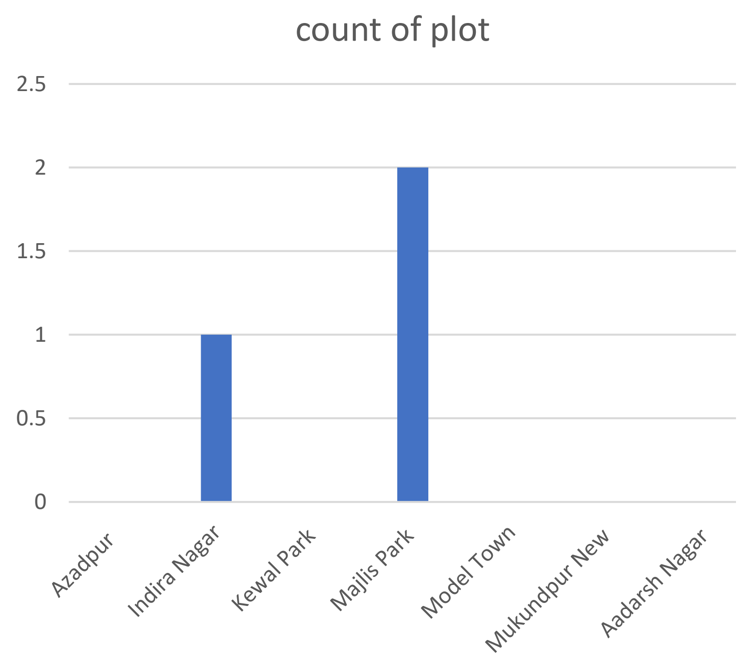


Figure 10:Bar graph showing count of plot

Analysis of Flats:

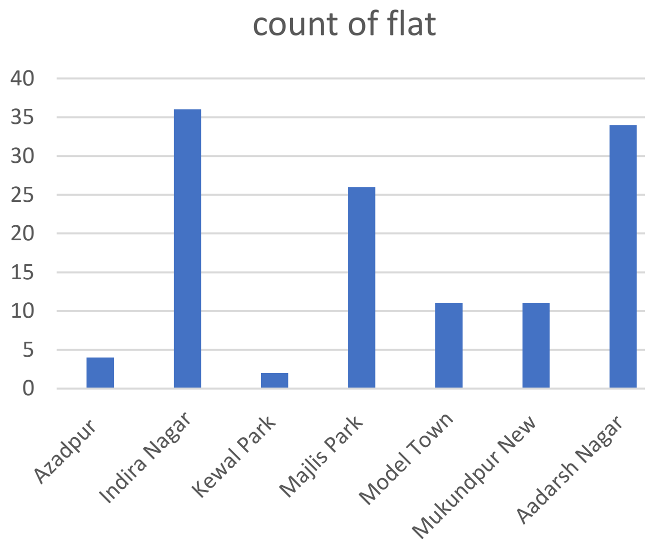


Figure 11: Bar graph showing count of flats

- **Indira Nagar** has the highest count of flats, implying it could be a popular residential area with a focus on apartments. This suggests demand for residential living spaces in this area is strong, potentially due to factors like infrastructure, location, or community preferences.

- **Adarsh Nagar** and **Majlis Park** also show a considerable count of flats, indicating they are significant residential zones.

- **Model Town** and **Mukundpur** are on par in terms of flat counts, showing moderate residential demand.

- **Azadpur** and **Kewal Park** have fewer flats, indicating these areas may be more commercially oriented or that there is less demand for apartment living in these locations.

Insights for Flats:

- Indira Nagar stands out as a primary residential market, suggesting strong demand for housing. This could be leveraged for residential development projects or marketing of flats.
- In areas like Azadpur, where the count of flats is low, there may be an opportunity to introduce residential projects if the area becomes more attractive for living spaces in the future.

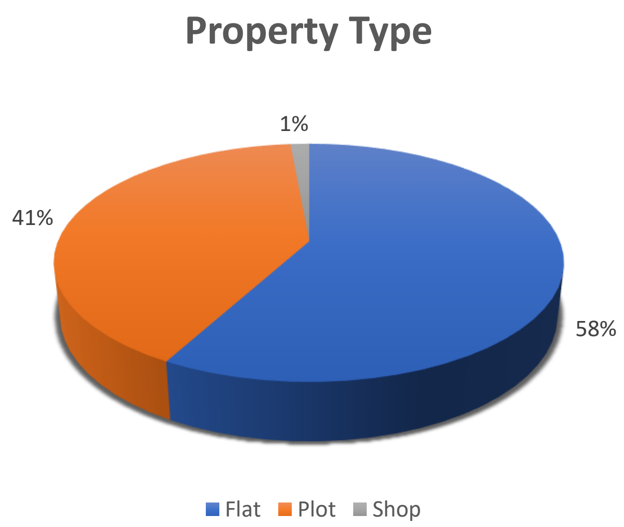


Figure 12:Pie chart showing proportions of each property types

Overall Insights Across Property Types:

- Azadpur is a dominant commercial hub, but less focused on residential (flats), meaning it may attract investors more interested in retail or commercial properties.
- Indira Nagar is a clear leader in the residential segment with a strong presence of flats, making it a key area for those focusing on housing markets.
- Model Town is balanced, showing strong commercial and moderate residential opportunities.
- Majlis Park stands out with the only available plots, though overall development in terms of flats and shops is lower.

Strategic Analysis and Recommendations:

- Commercial Focus: If company is more focused on commercial real estate, Azadpur and Model Town seem like optimal areas for focusing shop deals. Azadpur, in particular, could offer higher commissions given its dominance.

- Residential Focus: For residential property deals, Indira Nagar stands out as the key area for flat sales. Adarsh Nagar and Majlis Park could be secondary markets.

- Plot Development: Plots are scarce, especially outside of Majlis Park, which could suggest an opportunity for higher profit margins from rare land sales if we can secure and sell plots in these areas.

COUNT OF DEALS ANALYSIS:

Bar Graph Analysis (Count of Deals):

- **Rent (196 deals)** dominates the total number of transactions, compared to only **18 deals** for sale.
    - **Insight:** The real estate market your company is working in seems to be heavily focused on **rental** transactions rather than sales. This could indicate that the demand for renting properties is much higher, possibly due to affordability concerns, preference for short-term commitments, or the nature of the clientele (e.g., students, working professionals).
    - **Actionable Insight:** Business should prioritize rental properties in its strategy, focusing on maintaining a strong rental portfolio, which could lead to higher turnover and commissions. If business want to expand the sales segment, marketing efforts might need to focus on targeting buyers with offers or incentives to drive up sales.

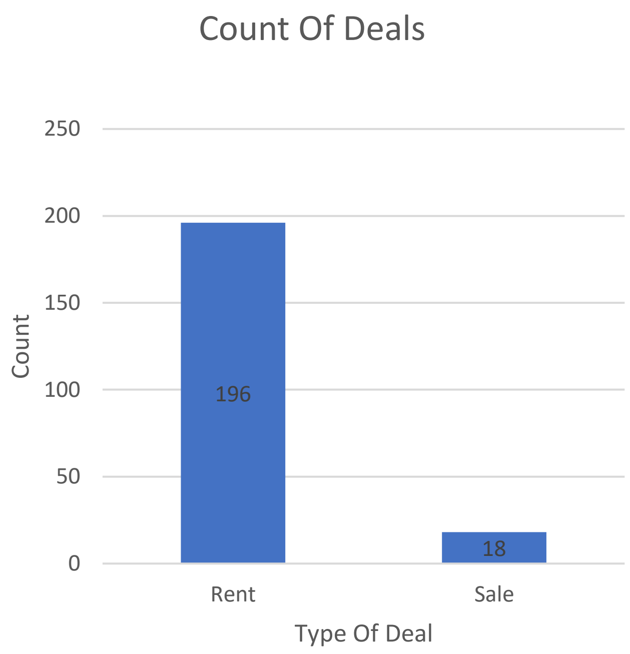


Figure 13:Count of rent and sale deals

Deal Type	total sum	Total commission earned from seller	Total commission earned from buyer	no of deals
Sale	₹24,55,00,000.00	₹19,55,000.00	₹22,55,000.00	197
Rent	₹46,53,000.00	₹23,26,500.00	₹23,26,500.00	19

Table Analysis (Sale and Rent):

Sales Segment:

- Total Sale Amount: ₹24,55,000
    - Compared to the rental market, the overall sum from sales is relatively low.

- Total Commission from Seller: ₹19,55,000, Total Commission from Buyer: ₹42,55,000
    - The commission earned from buyers is more than double that earned from sellers, indicating that buyers might be paying a premium or higher fees in this market.

    - Insight: Buyers in sales deals seem to contribute significantly more to the total commission compared to sellers, which may reflect the market dynamics where buyers are willing to invest more to secure a property. It could also indicate that more expensive or premium properties are sold than rented.

- Number of Deals (18): Sales are much fewer compared to rentals (only 18 deals).
    - Insight: This reinforces the earlier observation from the bar graph that the sales market is much smaller. The high commission from buyers in this segment might suggest targeting premium buyers could yield better returns

Rent Segment:

- Total Rent Amount: ₹46,53,000
    - Despite being a higher sum compared to the sales amount, this is expected given the high number of rental transactions.
- Total Commission from Seller: ₹43,26,500, Total Commission from Buyer: ₹43,26,500
    - Both the seller and buyer in rental deals contribute equally to the commission. This indicates a balanced approach in charging both parties for rental agreements.

    - Insight: Since commissions from rentals are substantial and nearly equal on both ends, there is significant revenue potential in the rental market, particularly given the high number of rental deals.

- Number of Deals (197): The high volume of rental deals drives most of the company's revenue.

    - Insight: Rental deals are not only more frequent, but they also generate significant commissions, making the rental market the core focus of business. This means business could continue expanding and fine-tuning its rental offerings, as this is the primary revenue source.

Lead Source Analysis

1. **Lead Volume**:
   - **Direct Leads (112)** are higher than **Personal Networking (102)**, indicating effective direct marketing strategies.

2. **Effectiveness**:
   - The high count from direct outreach suggests successful tactics. Analyzing these methods could optimize lead generation.

3. **Networking Strength**:
   - Personal networking is still significant. Consider enhancing efforts through events and referrals to increase lead conversion.

4. **Growth Potential**:
   - Both sources are close in numbers, indicating a balanced strategy is needed. Increasing networking activities could yield more leads.

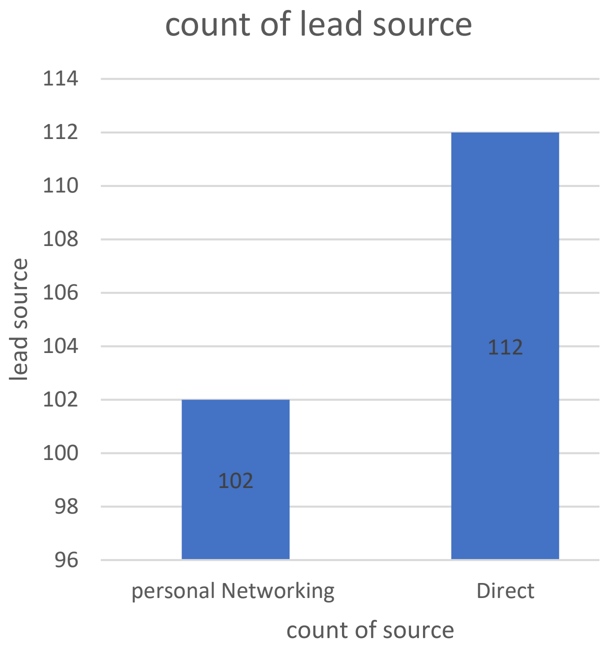


Figure 14: showing count of count from where mostly clients comes

Strategic Recommendations

1. **Enhance Direct Marketing**:
   - Focus on the most effective direct channels and experiment with messaging.

2. **Maximize Networking**:
   - Engage in local events and create referral programs to encourage client advocacy.

3. **Monitor and Measure**:
   - Track conversion rates from each source to assess effectiveness and adjust strategies accordingly.

4. **Integrate Efforts**:
   - Encourage network connections to advocate for your business and streamline follow-ups from networking leads.

DESCRIPTIVE MEASURES EXPLANATION FOR EACH LOCATION:

Descriptive Measures for Azadpur

Descriptive statistic measure	Amount of deal	Commission earned from buyer	Commission earned from seller	Expenses incurred	Renting	Sales
Sum	₹ 10,79,000.00	₹ 5,39,500.00	₹ 5,39,500.00	₹ 39,400.00	₹ 10,79,000.00	₹ 0.00
Mean	₹ 29,162.16	₹ 14,581.08	₹ 14,581.08	₹ 1,064.86	₹ 29,162.16	₹ 0.00
Standard Deviation	₹ 4,099.55	₹ 2,049.77	₹ 2,049.77	₹ 237.13	₹ 4,099.55	₹ 0.00
Minimum	₹ 17,000.00	₹ 8,500.00	₹ 8,500.00	₹ 700.00	₹ 17,000.00	₹ 0.00
Maximum	₹ 35,000.00	₹ 17,500.00	₹ 17,500.00	₹ 1,500.00	₹ 35,000.00	₹ 0.00

- Azadpur currently shows zero sales transactions, indicating a low demand for property purchases and suggesting a focus on rental strategies instead. The total rental deals amount to ₹10,79,000, with an average rent of ₹29,162, and a controlled price range of ₹17,000 to ₹35,000. Commission insights reveal a total of ₹5,39,500, averaging ₹14,581 per deal, with variability mostly between ₹8,500 and ₹17,500. Total expenses stand at ₹39,400, averaging ₹1,064 per deal, which remains consistent and low, contributing to profitability. Thus, all profits stem from rental deals, indicating that Azadpur is a profitable rental market. Moving forward, the focus should be on enhancing the quality and number of rental deals while reassessing sales strategies to understand the lack of property transactions in the area.

Descriptive Measures for Indira Nagar

Descriptive statistic measure	Amount of deal	Commission earned from buyer	Commission earned from seller	Expenses incurred	Renting	Sales
Sum	₹ 7,44,09,000.00	₹ 9,89,500.00	₹ 11,89,500.00	₹ 52,800.00	₹ 9,09,000.00	₹ 7,35,00,000.00
Mean	₹ 16,17,586.96	₹ 21,510.87	₹ 25,858.70	₹ 1,147.83	₹ 22,170.73	₹ 1,47,00,000.00
Standard Deviation	₹ 52,53,379.84	₹ 42,346.26	₹ 49,714.82	₹ 421.50	₹ 8,461.39	₹ 83,93,449.80
Minimum	₹ 15,000.00	₹ 0.00	₹ 7,500.00	₹ 700.00	₹ 15,000.00	₹ 35,00,000.00
Maximum	₹ 2,50,00,000.00	₹ 25,000.00	₹ 25,000.00	₹ 3,000.00	₹ 40,000.00	₹ 2,50,00,000.00

- The total value of property transactions is ₹7.44 crore, reflecting an active market with an average deal size of ₹16.18 lakhs, ranging from ₹15,000 to ₹2.5 crore. Commissions from buyers total ₹9.89 lakh (mean ₹21,510.87, range ₹0 to ₹25,000), while commissions from sellers total ₹11.89 lakh (mean ₹25,858.70, with all deals earning between ₹7,500 and ₹25,000). Total expenses are low and stable at ₹52,800 (average ₹1,147.83), simplifying budgeting. Renting revenue is ₹9.09 lakh (mean ₹22,170.73, range ₹15,000 to ₹40,000), but sales dominate at ₹7.35 crore (mean ₹1.47 crore, range ₹35 lakh to ₹2.5 crore). The market is driven by high-value sales across various property types (residential and commercial), yielding higher seller commissions and stable expenses that enhance profitability.

Descriptive Measures for Majlis Park

Descriptive statistic measure	Amount of deal	Commission earned from buyer	Commission earned from seller	Expenses incurred	Renting	Sales
Sum	₹ 6,20,55,000.00	₹ 8,92,500.00	₹ 6,42,500.00	₹ 33,400.00	₹ 5,55,000.00	₹ 6,15,00,000.00
Mean	₹ 1,88,04,454.54	₹ 27,045.45	₹ 19,469.69	₹ 1,012.12	₹ 19,137.93	₹ 1,53,75,000.00
Standard Deviation	₹ 66,83,865.19	₹ 64,546.29	₹ 50,759.22	₹ 403.66	₹ 5,026.53	₹ 1,41,50,235.50
Minimum	₹ 16,000.00	₹ 8,000.00	₹ 0.00	₹ 600.00	₹ 16,000.00	₹ 30,00,000.00
Maximum	₹ 3,00,00,000.00	₹ 3,00,000.00	₹ 3,00,000.00	₹ 2,500.00	₹ 35,000.00	₹ 3,00,00,000.00

- The total amount of property deals in Majlis Park is ₹6.2 crore, with a mean of ₹1.88 crore and a wide variation from ₹16,000 to ₹3 crore, indicating both small and high-value transactions. Commissions earned from buyers total ₹8.92 lakh (mean ₹27,045.45) with significant variability (min: ₹8,000, max: ₹3 lakh) linked to deal sizes, while seller commissions total ₹6.42 lakh (mean ₹19,469.69), also varying widely (min: ₹0, max: ₹3 lakh). Total expenses incurred are ₹33,400 (mean ₹1,012.12), showing consistent low costs (min: ₹600, max: ₹2,500) that enhance cost-efficiency. Renting contributes ₹5.55 lakh (mean ₹19,137.93) with moderate variability (min: ₹16,000, max: ₹35,000), but sales dominate with ₹6.15 crore (mean ₹1.53 crore) and high variability (min: ₹30 lakh, max: ₹3 crore). This sales-driven market presents significant opportunities for high-value transactions, variable commissions, and low expenses that boost profitability.

Descriptive Measures for Model Town

Descriptive statistic measure	Amount of deal	Commission earned from buyer	Commission earned from seller	Expenses incurred	Renting	Sales
Sum	₹ 4,52,07,000.00	₹ 10,43,500.00	₹ 7,93,500.00	₹ 49,000.00	₹ 12,07,000.00	₹ 4,40,00,000.00
Mean	₹ 11,02,609.75	₹ 25,451.21	₹ 19,353.65	₹ 1,195.12	₹ 31,763.15	₹ 14,66,66,667.00
Standard Deviation	₹ 45,17,177.56	₹ 41,950.23	₹ 21,848.41	₹ 346.19	₹ 9,104.30	₹ 1,05,03,967.50
Minimum	₹ 16,000.00	₹ 8,000.00	₹ 8,000.00	₹ 700.00	₹ 16,000.00	₹ 4,00,00,000.00
Maximum	₹ 2,50,00,000.00	₹ 25,000.00	₹ 15,000.00	₹ 2,500.00	₹ 40,000.00	₹ 2,50,00,000.00

- Model Town exhibits a vibrant real estate market with a total transaction amount of ₹4.52 crores, highlighting its active nature. The average deal size stands at ₹11.02 lakhs, with a substantial standard deviation of ₹45.17 lakhs, indicating a wide range of property values from ₹16,000 to ₹2.5 crores. Commission earned from buyers totals ₹10.43 lakhs, averaging ₹25,451.21, while seller commissions amount to ₹7.93 lakhs, averaging ₹19,353.65. The overall expenses are relatively low at ₹49,000, ensuring cost efficiency. The rental market is significant, with total rental income of ₹12.07 lakhs and an average rental price of ₹31,763.15. Sales transactions dominate with a total of ₹4.4 crores, reflecting the area's potential for high-value deals. The combination of high sales values, a robust rental market, and consistent commissions contributes to Model Town's overall profitability.

Descriptive Measures for Mukundpur

Descriptive statistic measure	Amount of deal	Commission earned from buyer	Commission earned from seller	Expenses incurred	Renting	Sales
Sum	₹ 3,01,99,000.00	₹ 3,99,500.00	₹ 3,99,500.00	₹ 12,700.00	₹ 1,99,000.00	₹ 3,00,00,000.00
Mean	₹ 25,16,583.33	₹ 33,291.66	₹ 33,291.66	₹ 1,058.33	₹ 18,090.90	₹ 3,00,00,000.00
Standard Deviation	₹ 82,86,562.12	₹ 83,996.87	₹ 83,996.87	₹ 467.50	₹ 1,928.47	₹ 0.00
Minimum	₹ 14,000.00	₹ 7,000.00	₹ 7,000.00	₹ 800.00	₹ 14,000.00	₹ 3,00,00,000.00
Maximum	₹ 3,00,00,000.00	₹ 3,00,000.00	₹ 3,00,000.00	₹ 2,500.00	₹ 19,000.00	₹ 3,00,00,000.00

- The extracted text contains descriptive statistics for Mukundpur, presenting various metrics across multiple categories. The total deal amount sums up to ₹3,01,99,000.00, while the commission earned from buyers totals ₹39,95,000.00. The commission earned from sellers is ₹3,99,500.00, and expenses incurred are ₹12,700.00. The rental income stands at ₹1,99,000.00, and sales total ₹3,00,00,000.00.

- The mean deal amount is ₹25,16,583.33, with buyer commissions averaging ₹33,291.66 and seller commissions also at ₹33,291.66. The expenses average at ₹1,058.33, while rental income averages ₹18,090.90, with sales mean at ₹3,00,00,000.00. The standard deviation for deal amounts is ₹82,86,562.12, indicating high variability, while commissions from buyers and sellers show a standard deviation of ₹83,996.87. The minimum deal amount is ₹14,000.00, with a maximum of ₹3,00,00,000.00. Overall, Mukundpur exhibits significant activity with varied deal sizes, consistent commissions, and stable expenses, suggesting a dynamic real estate market.

Descriptive Measures for Aadarsh Nagar

Descriptive statistic measure	Amount of deal	Commission earned from buyer	Commission earned from seller	Expenses incurred	Renting	Sales
Sum	₹ 2,91,80,000.00	₹ 6,25,000.00	₹ 5,25,000.00	₹ 50,600.00	₹ 6,80,000.00	₹ 2,85,00,000.00
Mean	₹ 7,11,707.31	₹ 1,54,243.90	₹ 13,125.00	₹ 1,234.14	₹ 17,894.73	₹ 95,00,000.00
Standard Deviation	₹ 28,12,898.96	₹ 26,271.68	₹ 22,783.75	₹ 1,265.03	₹ 6,310.82	₹ 57,66,281.29
Minimum	₹ 15,000.00	₹ 7,500.00	₹ 7,500.00	₹ 700.00	₹ 15,000.00	₹ 35,00,000.00
Maximum	₹ 1,50,00,000.00	₹ 9,000.00	₹ 1,50,000.00	₹ 1,50,000.00	₹ 35,000.00	₹ 1,50,00,000.00

- Aadarsh Nagar's real estate market shows a total deal amount of ₹2.91 crores, indicating a strong market, though slightly smaller than Mukundpur. The average deal size is lower, with a mean of ₹7,11,707 and a standard deviation of ₹28,12,899, reflecting large variation in deal sizes ranging from ₹15,000 to ₹1.5 crores. Buyer commissions total ₹6,25,000, resulting in a higher average of ₹1,54,243, with moderate variability (standard deviation of ₹26,271) and a range of ₹7,500 to ₹9,000. In contrast, seller commissions total ₹5,25,000, with a lower average of ₹13,125 and consistent variability (standard deviation of ₹22,784), ranging from ₹7,500 to ₹1.5 lakhs. Operational expenses are low at ₹50,600, with a mean of ₹1,234, but higher than those in Mukundpur, and vary between ₹700 and ₹1.5 lakhs. The rental market is healthy, totaling ₹6,80,000, with a mean rental income of ₹17,895 from properties ranging from ₹15,000 to ₹35,000. Sales dominate the market with a total of ₹2.85 crores and an average value of ₹95 lakhs, with deal sizes ranging from ₹35 lakhs to ₹1.5 crores. Overall, Aadarsh Nagar's market features a mix of small and large deals, high buyer commissions, manageable expenses, and a strong sales performance alongside mid-range rental properties.

Descriptive Measures for kewal Park

Descriptive statistic measure	Amount of deal	Commission earned from buyer	Commission earned from seller	Expenses incurred	Renting	Sales
Sum	₹ 80,24,000.00	₹ 92,000.00	₹ 92,000.00	₹ 4,800.00	₹ 24,000.00	₹ 8,00,000.00
Mean	₹ 20,06,000.00	₹ 23,000.00	₹ 23,000.00	₹ 1,200.00	₹ 12,000.00	₹ 40,00,000.00
Standard Deviation	₹ 24,42,958.86	₹ 21,260.29	₹ 21,260.29	₹ 673.30	₹ 0.00	₹ 14,14,213.56
Minimum	₹ 12,000.00	₹ 6,000.00	₹ 6,000.00	₹ 800.00	₹ 12,000.00	₹ 30,00,000.00
Maximum	₹ 50,00,000.00	₹ 50,000.00	₹ 50,000.00	₹ 2,200.00	₹ 12,000.00	₹ 50,00,000.00

- Kewal Park's real estate market exhibits a total deal amount of ₹80,24,000, with an average deal size of ₹20,06,000. The market shows significant variability in deal sizes, as evidenced by a standard deviation of ₹24,42,958.86, with deal amounts ranging from a minimum of ₹12,000 to a maximum of ₹50,00,000. The total commission earned from buyers is ₹92,000, with an average of ₹23,000, indicating moderate earnings from buyer transactions, while the standard deviation of ₹21,260.29 suggests some consistency in these commissions, with minimum and maximum values of ₹6,000 and ₹50,000, respectively. Seller commissions also total ₹92,000, mirroring buyer commissions, with the same average and standard deviation values, indicating that the earning potential from both sides is similar.

- Operational expenses in Kewal Park are low, totaling ₹4,800, with a mean expense of ₹1,200, reflecting a minimum of ₹800 and a maximum of ₹2,200. The rental market shows a total of ₹24,000 in rental income, with an average rental amount of ₹12,000, indicating steady but limited rental activity. Sales dominate the market, totaling ₹8,00,000, with an average of ₹40,00,000 and substantial variability, as indicated by a standard deviation of ₹14,14,213.56, with sales ranging from a minimum of ₹30,00,000 to a maximum of ₹50,00,000. Overall, Kewal Park presents a market with moderate deal sizes, consistent commissions, low operational expenses, and a strong emphasis on sales, while rentals appear to be less significant

COMPETITOR ANALYSIS:

In the competitive real estate market, a thorough competitor analysis is essential for strategic planning. We utilized Google Maps and Google search results to gather data on real estate agents in key areas such as Aadarsh Nagar, Azadpur, Kewal Park, Majlis Park, Model Town, and Mukundpur. This analysis reveals the number of agents listed, their ratings, and the services they offer.

Analysis and Insights

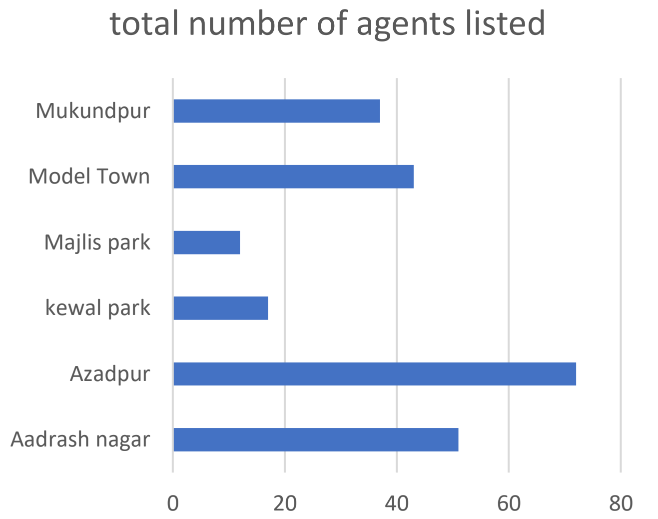


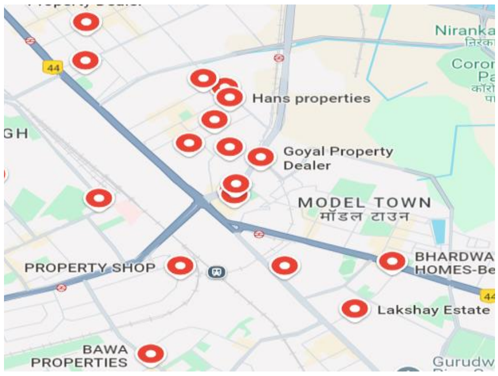
Figure 15:showing the count of competitors listed in each location

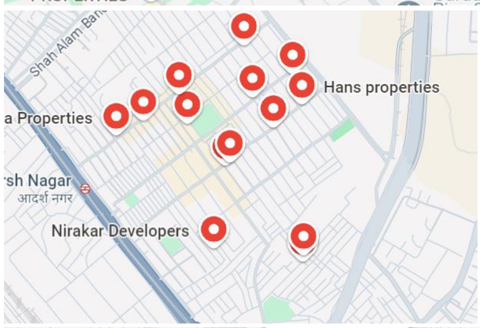
1. **Market Competition:**
   - **Azadpur** has the highest number of agents listed (72) with a good rating range (4.2-4.8), indicating a competitive market. This could suggest a higher demand for real estate services in that area, making it important for business to differentiate its offerings.
   - **Aadrash Nagar** also has a significant number of agents (51) with ratings between 4 and 4.5, which is a decent competition level.

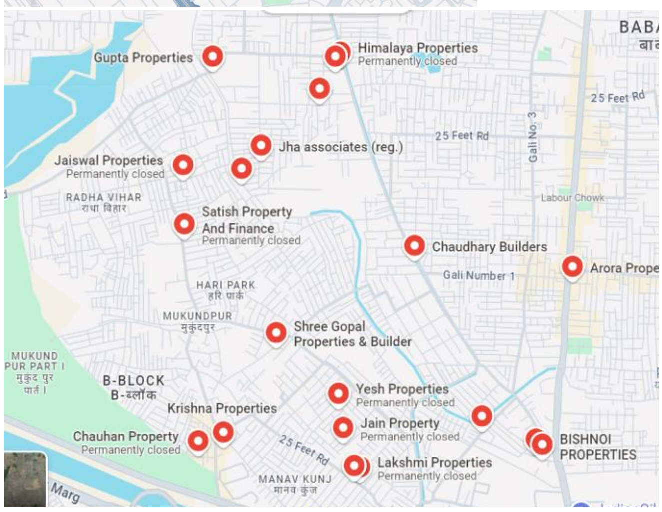
2. **Opportunity for Improvement:**
   - **Kewal Park** has the lowest number of agents listed (17) and a rating range of 3.8-4.2. This indicates less competition, which could be an opportunity for your business to capture a larger market share if you can improve business service offerings or market presence.
   - **Majlis Park** has only 12 agents with a rating range of 4-4.7, indicating a smaller market but a relatively good reputation, which could also be a target area for expansion.

3. **High-Quality Markets:**
   - **Model Town** stands out with the highest ratings (4.8-5) and a substantial number of agents (43). This area may attract more discerning clients who expect high-quality service, which could be an opportunity for premium offerings or specialized services.

4. **Strategic Focus:**
   - Areas like **Mukundpur** (37 agents, ratings 3-4.2) and **Kewal Park** (17 agents, ratings 3.8-4.2) may require a strategic approach focusing on improving customer satisfaction and service quality to compete effectively.







The insights can guide marketing strategies, such as targeting areas with fewer agents or enhancing your services in areas with higher competition but lower ratings to improve your standing.

Results and Recommendations

Primary Business Problem

The primary challenge Narang Associates faces is intense competition and lower profit margins in their current geographical areas. To overcome this, they can adopt the following strategic and marketing approaches:

Strategic Recommendations:

1. **Location-Specific Focus**
   - **Action:** Strengthen digital and traditional marketing efforts in **Indira Nagar** and **Majlis Park**. Offer special incentives to new buyers and maintain engagement with existing clients through personalized communications.
2. **Premium Focus for High-Value Locations**
   - **Action:** Focus on acquiring and promoting premium properties in **Adarsh Nagar** and **Model Town**. Cater to high-end clients by offering luxury property showcases and premium buyer services.
3. **Rental Strategy for Growing Areas**
   - **Action:** Position Narang Associates as the go-to expert for rental properties in **Model Town** and **Azadpur**. Partner with property owners to develop rental-specific listings and promote these aggressively through targeted marketing.
4. **Reevaluate Kewal Park**
   - **Action:** Explore niche markets such as affordable housing in **Kewal Park** or collaborate with local developers to boost market presence in this area.
5. **Balanced Portfolio Management**
   - **Action:** Maintain flexibility in locations like **Mukundpur** and **Adarsh Nagar**, where there's a mix of rental and sale properties. Promote multi-purpose properties that suit both renters and buyers, depending on demand.

Revenue and Growth Strategy

1. **Balanced Property Mix**
   - **Action:** Prioritize rental properties in rental hotspots and focus on premium properties in high-sales regions. Use data analysis to allocate resources based on future forecasts.
2. **Collaborative Efforts and Partnerships**
   - **Action:** Partner with developers to secure exclusive property rights or offer financing options, which can attract higher-value clients and improve overall sales.

Marketing Strategy for Narang Associates:

1. **Digital Marketing and Networking**
   - **Google Ads & SEO:** Invest in Google Ads and SEO to improve online visibility for high-value properties. Ensure listings appear on popular real estate portals such as MagicBricks and 99acres.
   - **Referral Programs:** Implement referral incentives for past clients, offering bonuses when their referrals result in a closed deal.
   - **Partnerships and Events:** Collaborate with local businesses and attend community events such as property expos and open houses to strengthen personal networking efforts.
   - **Impact:** Enhanced brand visibility, stronger client relationships, and more leads through online presence and personal connections.
2. **Advanced Marketing Techniques**
   - **Virtual Tours & 3D Walkthroughs:** Offer clients virtual property tours to make your service more accessible and stand out in the competitive digital marketplace.

- **Influencer Marketing:** Collaborate with real estate influencers who cater to buyers or investors on social media platforms.
- **Impact:** Improved client engagement and a broader audience reach, especially in the premium market.

Geographical Expansion Strategy:

1. **Target Upcoming Localities**
   - **Action:** Research and invest in areas with upcoming infrastructure developments, such as **Dwarka**, **Noida Extension**, and parts of **South Delhi** (e.g., **Saket**, **Vasant Kunj**). Focus on purchasing properties at the early stages of development to capitalize on rising property values.
   - **Impact:** Higher returns through strategic investments in high-growth areas.
2. **Collaborate with Developers for High-End Projects**
   - **Action:** Partner with builders who are developing high-end properties like luxury residential towers or premium commercial spaces. This allows Narang Associates to cater to affluent clients and increase commissions.
   - **Impact:** Establish Narang Associates as a premium service provider and differentiate from competitors by focusing on high-end projects.

Efficient Handling of Competition:

1. **Building Relationships with Other Agents**
   - **Action:** Collaborate with other real estate agents, allowing for shared listings or buyers and expanding market access.
   - **Impact:** Reduces rivalry and increases the client base through strategic partnerships.
2. **Client Retention Strategies**
   - **Action:** Offer loyalty programs, discounted legal services, or property management for long-term clients to improve retention.
   - **Impact:** A long-term relationship with clients increases recurring business and reduces acquisition costs.

Collaborate with Builders for New Projects:

1. **Joint Ventures with Builders**
   - **Action:** If Narang Associates is considering purchasing land to build multi-floor projects, collaborate with builders to share costs and reduce the financial burden. Secure clear legal agreements for profit-sharing and roles.
   - **Impact:** Lowered development costs and faster project completion.
2. **Pre-Construction Deals**
   - **Action:** Offer early-bird deals for properties under construction, attracting buyers with lower prices and generating interest in new listings.
   - **Impact:** Differentiation from competitors and early revenue generation from pre-construction investments.

Develop a Unique Selling Proposition (USP):

1. **Exclusive Listings**
   - **Action:** Partner with builders to secure exclusive marketing rights for under-construction projects. This gives Narang Associates a competitive edge in offering properties not available through other agents.
   - **Impact:** A unique offering draws clients to Narang Associates, enhancing market differentiation.
2. **Client-Centric Approach**
   - **Action:** Provide highly personalized services such as detailed property reports, financing guidance, and legal assistance.
   - **Impact:** Increases client trust, boosts referrals, and strengthens customer loyalty.

By implementing these strategies, **Narang Associates** can successfully navigate competition, expand into new markets, and increase profitability through well-defined marketing, partnerships, and geographical expansion. These recommendations will help the business grow sustainably while retaining existing clients and attracting higher-value prospects.